

Algorithm 1 OmegaGenome Two-Stage Training Workflow

Stage 1: Fine-tune Teacher(s)

Input: Data $\mathcal{D}_\tau$, Teachers $\mathcal{T}$

Output: Fine-tuned T_τ^* , Cache $\mathcal{C}$

```
1: for each task  $\tau$  do
2:   Select best teacher  $T \in \mathcal{T}$ 
3:   if  $T$  is Large (e.g., NT-2.5B) then
4:     LoRA: Freeze base, train adapter
5:   else
6:     Full: Fine-tune all params on  $\mathcal{D}_\tau$ 
7:   end if
8:   Set  $T_\tau^* \leftarrow T$ 
9:   Run:  $z^{(T)}, h^{(T)} \leftarrow T_\tau^*(x)$  for all  $x$ 
10:  Cache: Store logits  $z^{(T)}$  & feats  $h^{(T)}$ 
11: end for
```

Stage 2: Distill to Student

Input: Hypers α, β, γ , Temp τ , Cache $\mathcal{C}$

Output: Student S_τ^*

```
1: for each task  $\tau$  do
2:   Init Student  $S(\theta_S)$ 
3:   for epoch  $1 \dots E$  do
4:     for batch  $\mathcal{B} \subset \mathcal{D}_\tau$  do
5:       Get  $z^{(S)}, h^{(S)}$  from  $S(x; \theta_S)$ 
6:       Load  $z^{(T)}, h^{(T)}$  from Cache  $\mathcal{C}$ 
7:       CE:  $\mathcal{L}_{\text{CE}} \leftarrow \text{CrossEnt}(z^{(S)}, y)$ 
8:       KD:  $\mathcal{L}_{\text{KD}} \leftarrow \text{KL}(\sigma(\frac{z^{(T)}}{\tau}) \parallel \sigma(\frac{z^{(S)}}{\tau}))$ 
9:       Feat:  $\mathcal{L}_{\text{feat}} \leftarrow \|h^{(T)} - Wh^{(S)}\|_2^2$ 
10:      Total:  $\mathcal{L} \leftarrow \mathcal{L}_{\text{KD}} + \beta\mathcal{L}_{\text{CE}} + \gamma\mathcal{L}_{\text{feat}}$ 
11:      Step:  $\theta_S \leftarrow \text{AdamW}(\nabla \mathcal{L})$ 
12:    end for
13:  end for
14:  Return  $S_\tau^*$ 
15: end for
```